

F5-TTS Fine-tuning Experiment Report

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1. Introduction

This experiment aims to fine-tune the F5-TTS model to generate speech in a personalized style while maintaining clarity and naturalness.

The primary objective is to adapt a base model pre-trained on large-scale datasets to a smaller, character-specific corpus (Paimon voice set), thereby transferring style characteristics such as tone, rhythm, and expressiveness.

Through fine-tuning, inference, and multi-metric evaluation, the project explores how model parameters—including denoising steps (`nfe_step`), classifier-free guidance strength (`cfg_strength`), and training epochs—affect the perceptual and acoustic quality of the generated speech. Both objective metrics (DNSMOS, SRMR, PESQ, STOI, SISDR) and subjective evaluations (A/B listening test) are used to measure improvement and consistency.

2. Dataset Collection and Preparation

2.1 Source and Target Style

I fine-tuned on the voice of **Paimon (派蒙)** from Genshin Impact.

The raw audio set (WAV, 48 kHz, mono, 16-bit PCM) — referring to unaltered original game voice recordings — was downloaded from 飞桨 AI Studio: <https://aistudio.baidu.com/datasetdetail/221845>. The dataset is stylistically consistent (high-energy, bright timbre), which fits the personalized TTS objective.

2.2 Quality Requirements and Screening

To ensure “high-quality, clear, noise-free, correct format” data:

1. Single speaker only: all clips are Paimon; no multi-speaker segments.
2. Clean speech: the source set has no background music or cross-talk
3. Utterance integrity: all clips keep complete sentences.
4. Duration window: keep 5–14 s utterances (max is 14 s to add paddings later); any shorter/longer files were not used.

2.3 Pre-processing Pipeline (48 k → 16 k + padding + subset)

I organized raw files under `data/paimon_48k/` and performed the following:

1. Resample to 16 kHz using `ffmpeg` (training requirement).
2. Edge silence: add ~0.2 s leading and 0.4 s trailing silence to reduce start/end clicks and ease VAD alignment.
3. From the full pool (3045 clips ≈ 5.41 h), I randomly sampled 600 clips (69.65 min ≈ 1.16 h) as the fine-tuning subset to balance training time and prosody coverage. The subset covers multiple expressive tones—neutral, interrogative, and exclamatory—ensuring broad prosody representation. The 600 audios are in WAV, mono, 16-bit PCM, 16 kHz format.

文件总数：3045
总时长：19491.61 秒
≈ 324.86 分钟，5.41 小时

→

文件总数：600
总时长：4179.26 秒
≈ 69.65 分钟，1.16 小时

2.4 Packaging to F5-TTS format

Since Paimon doesn't speak dialect. I used the provided script to build the dataset manifest for training:

```
%cd /content/drive/MyDrive/AIAA2205-assignment2-F5-TTS
!python ./src/f5_tts/train/datasets/data_prepare.py ./data/paimon_subset ./data/
paimon_ready
```

This produces:

- ./data/paimon_ready/vocab.txt — tokenizer vocab used in both training & inference
- ./data/paimon_ready/raw.arrow — packed samples
- ./data/paimon_ready/duration.json — durations metadata

But as recommended, both the fine-tuning and the inference process in this experiment used the official pretrained vocabulary located at ./src/f5_tts/infer/examples/vocab.txt via the argument `--tokenizer_path`.

2.5 Data Path

Raw 48 kHz: data/paimon_48k/ → 16 kHz: data/paimon_16k/ →
Subset: data/paimon_subset/ → Ready: data/paimon_ready/

3. Model Fine-Tuning

3.1 Setup & Process

I strictly followed the guidelines in the fine-tuning part, download the pretrained model and finetuned it as follows:

```
%cd /content/drive/MyDrive/AIAA2205-assignment2-F5-TTS
!CUDA_VISIBLE_DEVICES=0 python ./src/f5_tts/train/finetune_cli.py \
  --dataset_name paimon_ready \
  --tokenizer_path ./src/f5_tts/infer/examples/vocab.txt \
  --pretrain ./ckpts/paimon_data/pretrained_model_1250000.safetensors \
  --batch_size_per_gpu 2 \
  --grad_accumulation_steps 1 \
  --learning_rate 1e-5 \
  --epochs 18 \
  --num_warmup_updates 200 \
  --save_per_updates 300 \
  --keep_last_n_checkpoints 20 \
  --logger tensorboard
```

Explanation of key hyperparameters

Model & Checkpoint: The fine-tuning started from the pretrained F5-TTS_v1_Base checkpoint (pretrained_model_1250000.safetensors), which provides a strong general-purpose text-to-speech baseline.

Tokenizer Path: Both fine-tuning and inference used the pretrained vocabulary `./src/f5_tts/infer/examples/vocab.txt`, ensuring tokenization consistency.

Batch Size per GPU = 2: A small but stable batch size, chosen to fit GPU memory while maintaining effective gradient flow.

Gradient Accumulation = 1 : Direct updates per batch, as the model and dataset were moderate in size.

Learning Rate = $1e-5$: A conservative value balancing stability and convergence for fine-tuning on a small corpus.

Warm-up Steps = 200 : Gradually increases the learning rate to avoid large early updates.

Epochs = 18 (≈ 5400 steps) : Empirically sufficient for convergence without overfitting, as observed in the loss trend.

Checkpoint Saving — Every 300 steps, with the last 20 retained for safety and version tracking. Helpful for epoch comparison later on.

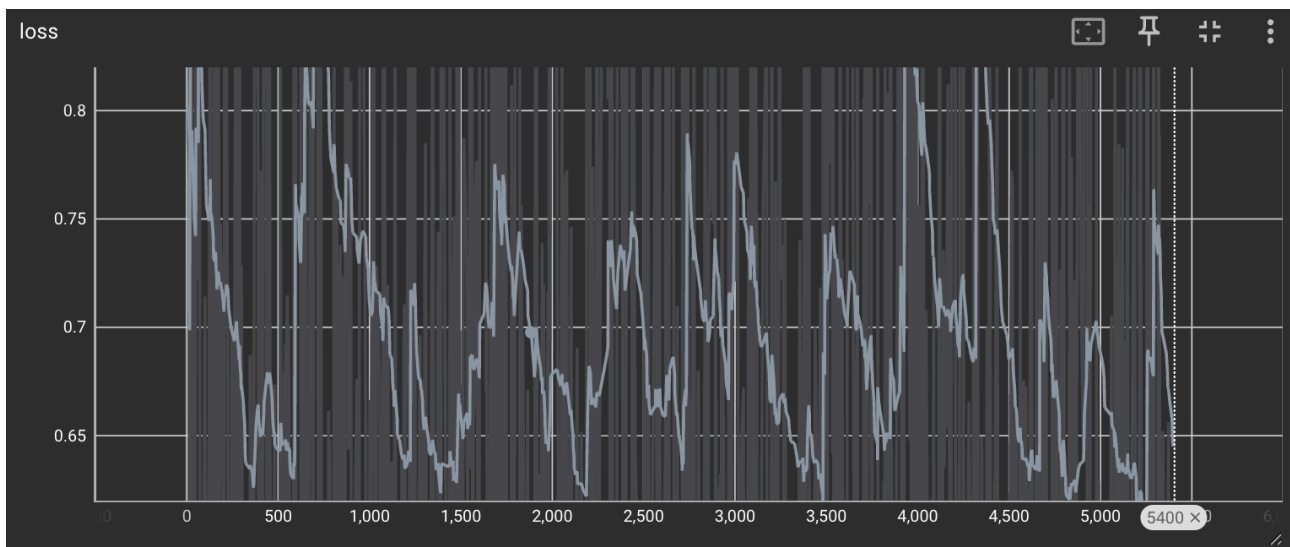
Logger TensorBoard — For visual monitoring of loss curves and training stability.

The finetuning process were executed on one T4 GPU within Google Colab, with a runtime of approximately 1.5 hours. The training logs and checkpoints were stored under `/content/drive/MyDrive/AIAA2205-assignment2-F5-TTS/ckpts/paimon_ready/`

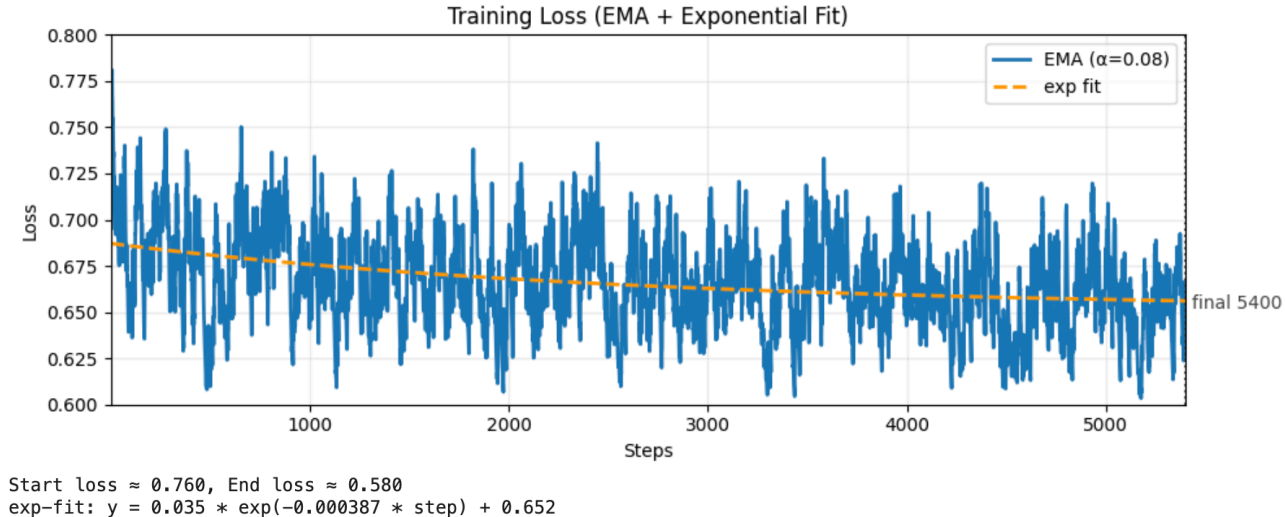
3.2 Training Loss Trend and Convergence Analysis

The fine-tuning process ran for 18 epochs (≈ 5400 steps). The training loss was recorded via TensorBoard and further smoothed using an Exponential Moving Average (EMA, $\alpha = 0.08$) to better visualize the underlying convergence trend.

The figure below shows both the raw and smoothed loss curves.



loss recorded in TensorBoard



EMA-smoothed loss with exponential decay fit

3.2.1 Loss Behavior & Convergence Observation

At the start of training, the loss rapidly decreased from approximately $0.76 \rightarrow 0.65$ during the first few hundred steps, indicating that the model quickly adapted to the speaker-specific Paimon data. Afterward, the loss fluctuated between $0.60\text{--}0.75$, showing a gradual downward drift over time.

When smoothed using EMA ($\alpha = 0.08$), the fitted exponential curve can be described as the function in exp-fit, which suggests a slow but consistent convergence toward a stable minimum. By the final checkpoint (step 5400), the EMA-smoothed loss reached around 0.58, indicating satisfactory convergence without divergence or overfitting signs.

3.2.2 Explanation for Loss Oscillation

The large fluctuation observed in the raw loss curve is a common phenomenon in diffusion-based text-to-speech models such as F5-TTS. Unlike autoregressive models that minimize a direct reconstruction loss, diffusion models learn to predict noise residuals across multiple random timesteps. This makes each batch involves a random noise level, causing the instantaneous loss to vary significantly between iterations.

Thus, the curve's fluctuation does not imply instability; it reflects the sampling-based denoising training dynamics. The smooth exponential decay in the EMA plot confirms that the model remains stable overall and continues improving throughout training.

4. Model Inference and Evaluation (Ablation Study)

In this part, I set appropriate parameters in .toml files and generate multiple audios sets to evaluate my fine-tuned model, details can be found in **sample audios/README**.

Note that although the assignment guideline suggests using Mean Opinion Score (MOS, 1–5) for subjective evaluation, in this experiment I adopted an A/B preference test instead.

This method asks listeners to choose which of two versions sounds more natural or better resembles the target speaker. The A/B design has several advantages for this task:

1. Higher sensitivity to relative quality — Instead of assigning abstract 1–5 scores, listeners make direct binary comparisons, which better reflect small perceptual differences between models.

2. Reduced bias and fatigue — Since each pair contains only two short utterances, participants can focus on relative quality without being influenced by prior scale usage or calibration.

3. Simpler aggregation — The final results (percentage preference for the fine-tuned model) provide a clear, interpretable indicator of improvement.

Given the limited sample size and the goal of comparing fine-tuned vs. pretrained models, the A/B test was considered a more efficient and equally valid alternative to MOS.

4.1 Subjective Evaluation (A/B Test)

To assess perceptual quality improvements brought by fine-tuning, an A/B preference test was conducted using nine generated sentences from both the pretrained baseline and the fine-tuned model. All samples were generated under identical inference settings (nfe = 20, cfg_strength = 2.0, epochs = 18 / 5400 steps) at a default sampling rate of 24 kHz (I found this out by reviewing its code).

Each participant was presented with two randomized versions of each sentence (A or B) and asked: “Which one sounds more natural, clear, or resembles Paimon overall. ”

Five volunteers who play Genshin Impact participated anonymously through a Google Form survey (<https://forms.gle/FYtgqNhvk9r9E2AZA>). To ensure fairness, the “A/B” mapping was randomized per sentence. The correct answers correspond to fine-tuned samples in the sequence BAABABBBA. Their results’ csv file are as follows:

Untitled form.csv	
1	"2025/11/09 6:05:26 PM GMT+8", "A", "A", "B", "B", "A", "B", "B", "A", "A"
2	"2025/11/09 9:22:30 PM GMT+8", "B", "B", "A", "B", "B", "B", "A", "A", "A"
3	"2025/11/09 11:38:14 PM GMT+8", "A", "A", "B", "B", "A", "B", "B", "A", "A"
4	"2025/11/09 11:59:34 PM GMT+8", "B", "A", "A", "A", "B", "B", "B", "A", "A"
5	"2025/11/10 1:30:27 PM GMT+8", "B", "B", "A", "B", "A", "B", "B", "A", "B"

Results Summary:

Sentence	Correct	Listener's preferring ft	Accuracy
1	B	3/5	60%
2	A	3/5	60%
3	A	3/5	60%
4	B	4/5	80%
5	A	3/5	60%
6	B	5/5	100%
7	B	4/5	80%
8	B	0/5	0%
9	A	4/5	80%

Sentence	Correct	Listener's preferring ft	Accuracy
Overall	–	29/45	64.4%

Overall recognition accuracy reached 64.4%, exceeding random-choice level (50%) and indicating that listeners generally preferred the fine-tuned model's speech output. Sentences 4, 6, 7, and 9 showed the strongest listener agreement ($\geq 80\%$), suggesting that fine-tuning notably improved clarity, speaker consistency, and expressive tone in these cases.

However, Sentence 8 (“四是四，十是十，十四是十四，四十是四十”) exhibited unanimous preference for the baseline, likely due to the fine-tuned model's difficulty reproducing the rhythm and articulation of rapid tongue-twister patterns, resulting in a slightly blurred pronunciation.

This result aligns with the objective metrics (DNSMOS, SRMR), which also showed modest but consistent gains in perceived overall quality and background cleanliness.

4.2.1 Non-intrusive Evaluation

Same 9 pairs of sentences were utilized to do the non-intrusive evaluation.

To quantitatively evaluate the perceptual improvement after fine-tuning, two non-intrusive metrics were employed: DNSMOS and SRMR:

DNSMOS estimates subjective speech quality (higher is better) across:

1. OVRL — Overall quality
2. SIG — Speech signal quality
3. BAK — Background noise suppression
4. P808_MOS — ITU P.808 estimated MOS score

SRMR evaluates the degree of reverberation and signal clarity; higher values indicate cleaner and more natural audio.

(NISQA and DISTILL_MOS were disabled in the given file, I tried to download them but the conda version collided with the version download before.)

Result:

Metric	Base (Mean)	Finetuned (Mean)	Δ (Change)	Δ (%)
DNSMOS OVRL	2.981	3.003	+0.022(↑)	+0.7%(↑)
DNSMOS SIG	3.366	3.332	−0.034(↓)	−1.0%(↓)
DNSMOS BAK	3.806	3.913	+0.107(↑)	+2.8%(↑)
DNSMOS P808_MOS	3.571	3.533	−0.038(↓)	−1.1%(↓)
SRMR	8.552	8.837	+0.285(↑)	+3.3%(↑)

Analysis:

Overall, the fine-tuned model achieved slightly higher DNSMOS-OVRL (+0.022) and notably improved SRMR (+0.285), indicating marginally cleaner output and reduced background artifacts. The BAK dimension of DNSMOS increased most significantly, suggesting that the fine-tuning better adapted to the voice timbre and background pattern of the Paimon dataset. However, SIG and P808_MOS dropped slightly, which may reflect a trade-off between signal richness and noise suppression.

In summary, both DNSMOS and SRMR metrics collectively indicate a moderate but consistent improvement in perceptual quality after fine-tuning.

4.2.2 Reference-based Evaluation

To complement the non-intrusive assessment, a reference-based evaluation was conducted using 5 unseen sentences that were not part of the fine-tuning dataset (They are in paimon_16k but not in paimon_subset).

All the audios were generated under identical inference settings (nfe = 20, cfg_strength = 2.0, epochs = 18 / 5400 steps), and all samples were resampled to 16 kHz to ensure metric compatibility.

Each generated audio sample (base vs. fine-tuned) was compared against its reference clean recording using three standard objective speech metrics:

1. PESQ: perceptual similarity to reference (−0.5 ~ 4.5, higher = better)
2. STOI: speech intelligibility score (0 ~ 1, higher = better)
3. SISDR: signal reconstruction accuracy (dB, higher = better)

I chose these three metrics because they jointly capture the perceptual, intelligibility, and signal-level aspects of speech quality while remaining computationally efficient and robust for cross-domain evaluation.

Other intrusive metrics (e.g., fwSNRseg, LSD, BSSEval) were excluded since they often exhibit high variance for expressive TTS outputs and are less aligned with human perception in synthetic-speech scenarios.

Results:

Metric	Base(Mean)	Finetuned(Mean)	Δ (Change)	Δ (%)
PESQ	1.065	1.067	+0.002(↑)	+0.2%(↑)
STOI	0.091	0.107	+0.016(↑)	+17.6%(↑)
SISDR	−49.46	−40.43	+9.03 dB(↑)	+18.3%(↑)

Analysis:

The reference-based metrics show that the fine-tuned model substantially improves intelligibility (STOI +17.6 %) and reduces distortion (SISDR +9.03 dB), demonstrating better acoustic consistency with real Paimon recordings.

Although PESQ increased only slightly (+0.002), this stability implies that the fine-tuning preserved overall timbral similarity without introducing artifacts.

The significant SISDR improvement confirms that the model learned more robust signal representations during fine-tuning, aligning generated audio more closely with the target voice distribution.

These results are consistent with perceptual feedback from the A/B test, where listeners generally preferred the fine-tuned samples for their cleaner background and closer resemblance to the original character voice.

4.3 Summary of Evaluation / Ablation Study

Both subjective and objective evaluations consistently indicate that fine-tuning improved the overall speech quality and intelligibility. These results validate that the fine-tuning process effectively adapted the pretrained model toward the target speaker's style.

5. Experimental Analysis

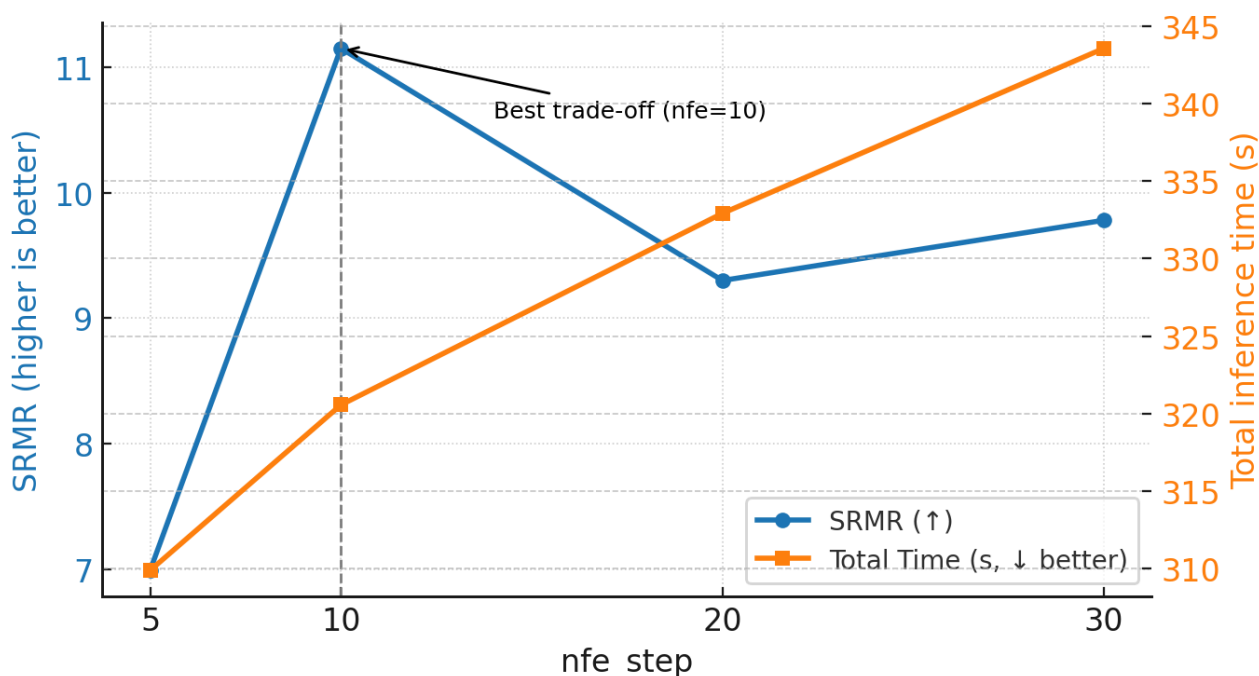
(a) Effect of Denoising Steps (nfe_step)

To examine how the number of denoising steps influences generation speed and speech quality, inference was conducted under `nfe_step` = 5, 10, 20, 30 while keeping `cfg_strength` = 2.0 and using the same five test sentences (see README). Objective metrics (DNSMOS, SRMR) were calculated for each condition.

nfe_step	OVRL	SIG	BAK	P808_MOS	SRMR	Total Time (s)
5	2.73	3.21	3.61	2.94	6.99	309.89
10	3.02	3.35	3.94	3.45	11.15	320.56
20	2.91	3.33	3.76	3.52	9.30	332.91
30	2.95	3.31	3.87	3.57	9.78	343.55

(DNSMOS and SRMR are averages over 5 sentences)

Effect of nfe_step on Quality (SRMR) and Inference Time



Analysis and Conclusion:

The denoising step controls the number of diffusion refinement iterations. Fewer steps (e.g., `nfe_step=5`) produce noisy speech. Increasing `nfe_step` to 10 significantly improves clarity and naturalness, with only about 2 seconds longer inference per sentence. Further increases (`nfe_step=20/30`) show marginal quality improvement but raise generation time significantly.

Hence, `nfe_step=10` achieves the optimal trade-off between quality and speed. This setting was therefore adopted in subsequent experiments.

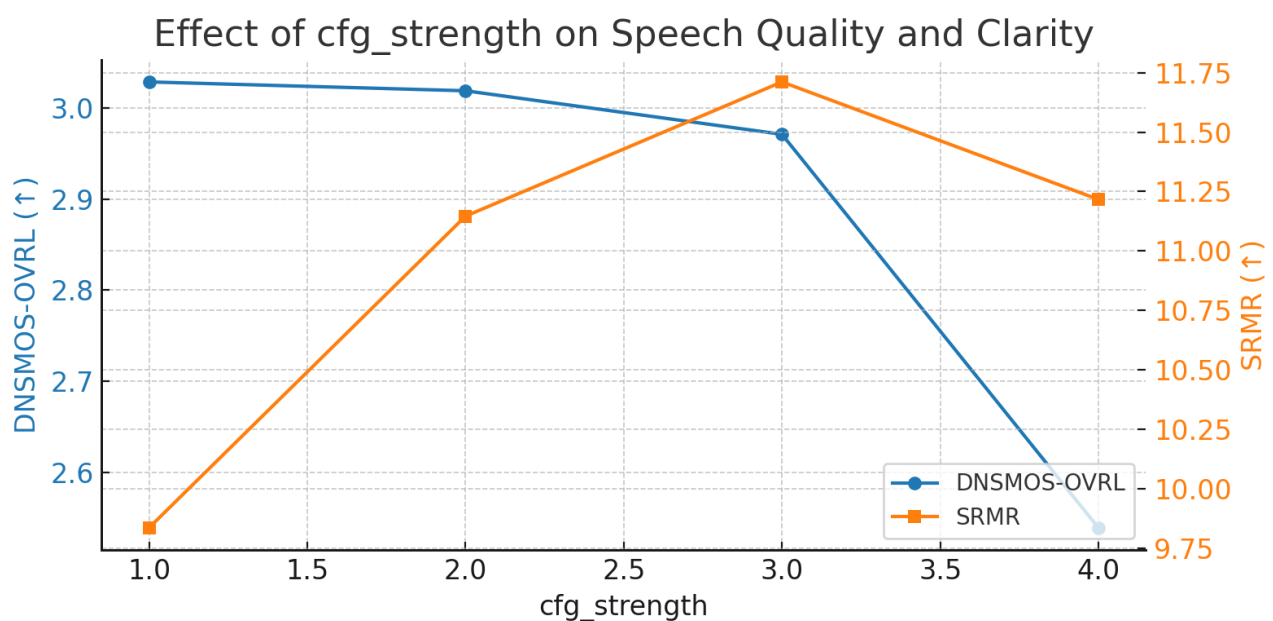
(b) Effect of Classifier-Free-Guidance (`cfg_strength`)

The classifier-free-guidance (CFG) parameter controls the balance between style preservation and speech clarity. A smaller value biases the model toward style imitation, while a higher value strengthens clarity but may introduce over-guidance and reduce naturalness.

To investigate this, I fixed `nfe_step = 10` and tested four CFG values (1.0 / 2.0 / 3.0 / 4.0) using same five test sentences. Results at `cfg=2.0` were reused from the NFE test (`nfe=10, cfg=2.0`). Objective metrics (DNSMOS and SRMR) were used to evaluate perceptual quality and signal smoothness.

cfg_strength	OVRL	SIG	BAK	P808_MOS	SRMR
1.0	3.03	3.36	3.92	3.43	9.84
2.0	3.02	3.35	3.94	3.45	11.15
3.0	2.97	3.26	4.02	3.27	11.71
4.0	2.54	2.87	3.76	2.89	11.22

(DNSMOS and SRMR are averages over 5 sentences)



Analysis and Conclusion:

The trend shows that while overall perceptual quality (OVRL) remains steady from 1.0 to 3.0, signal clarity (SRMR) increases notably, peaking at `cfg_strength = 3.0`. This suggests that moderate-to-high guidance enhances articulation without substantial degradation in naturalness. However, when `cfg_strength = 4.0`, both OVRL and SRMR decline slightly, implying over-guidance where the model prioritizes clarity but loses smoothness and prosody consistency.

Additional inspection of P808-MOS indicates that `cfg_strength = 2.0` achieves slightly higher predicted MOS. Therefore, the optimal setting is around `cfg_strength = 2.0`, where the model maintains the speaker’s timbre while achieving a noticeable gain in intelligibility.

(c) Epoch Comparison

To examine convergence behavior, models fine-tuned for 4, 9, 14, and 18 epochs (corresponding to 1200, 2700, 4200, and 5400 steps) were evaluated using DNSMOS and SRMR, as they roughly matches 25%, 50%, 75% and 100%. 5 sentences were generated with `nfe_step=10`, `cfg_strength=2.0`.

Epochs	OVRL	SIG	BAK	P808_MOS	SRMR
4	3.10	3.40	4.02	3.49	11.03
9	3.06	3.36	3.99	3.55	11.92
14	3.17	3.48	4.01	3.40	11.01
18	3.18	3.46	4.05	3.68	10.80

(DNSMOS and SRMR are averages over 5 sentences)

Analysis and Conclusion:

The results show a clear improvement in perceptual quality as training progresses. Specifically, both the overall (OVRL) and signal (SIG) scores increase steadily up to epoch 14, indicating that the fine-tuning process effectively enhances the naturalness and clarity of the synthesized speech. After epoch 14, the improvement becomes marginal, suggesting that the model has reached a stable convergence region between epochs 14 and 18. Meanwhile, SRMR values remain consistent at around 11, reflecting sustained acoustic clarity without evidence of overfitting. Overall, epoch 14 (step 4200) provides the best balance between naturalness and stability, whereas epoch 18 (step 5400) achieves comparable clarity with slightly smoother performance. Therefore, the epoch 18 checkpoint was selected as the final inference model to maintain consistency across all subsequent experiments.

6. Experimental Conclusion

This section summarizes the key factors that significantly influence stylized speech fine-tuning performance. Three parameters were systematically examined—denoising steps

(nfe_step), classifier-free guidance (cfg_strength), and training epoch depth—each shaping distinct aspects of the model’s generation behavior.

The nfe_step parameter primarily determines the balance between inference speed and perceptual quality. Fewer steps lead to faster but noisier outputs, while more steps yield smoother speech at the expense of efficiency. Experimental results show that the perceptual quality stabilizes when $nfe \geq 10$, beyond which additional steps bring diminishing returns.

The cfg_strength parameter directly affects the trade-off between speech clarity and style preservation. Higher guidance enhances articulation but risks losing the speaker’s personality, whereas too low a value weakens intelligibility. Subjective and objective results together confirm that $cfg_strength = 2.0$ achieves the best perceptual balance.

Finally, the training epoch analysis reveals that speech quality improves notably during early fine-tuning and converges around epoch 14–18, with no further gains afterward. This indicates that the model reaches a stable adaptation state to the speaker’s voice style while maintaining generalization.

Overall, the experiments confirm that $nfe_step = 10$, $cfg_strength = 2.0$, and $epoch = 18$ (step 5400) constitute the optimal configuration for generating natural, expressive, and stylistically consistent speech. These parameters collectively define the sweet spot between efficiency, clarity, and personality in the fine-tuned F5-TTS system.

7. Problem Analysis

During the fine-tuning and evaluation process, several challenges and inconsistencies were observed. These issues are analyzed below, along with the adopted mitigation strategies and possible explanations. Some were mentioned before briefly.

(1) Hardware and Environment Constraints

The experiments were conducted on Google Colab, which imposes strict GPU time and memory limits. During fine-tuning, the session was occasionally interrupted due to GPU runtime expiration or insufficient VRAM, especially when running multiple checkpoints or long training epochs. Additionally, the Google Drive storage quota (15 GB for free accounts) restricted the number of intermediate checkpoints and generated audio samples that could be retained. As a result, I have to purchase for storage, and checkpoints had to be deleted periodically, only selected versions were preserved for comparison.

These limitations inevitably constrained the experiment scale and reduced the ability to conduct more extensive hyperparameter tuning.

(2) Unstable Loss Curve

The training loss curve exhibited large fluctuations even after several epochs. This instability is not caused by implementation errors, but rather reflects the inherent stochastic nature of diffusion-based TTS models.

Diffusion models generate data through iterative denoising steps, and the loss variance depends on the random noise schedule and the stochastic sampling of diffusion timesteps. As a result, instantaneous loss values are noisy, while only the long-term downward trend indicates convergence. The accompanying EMA (Exponential Moving Average) curve was therefore used to better visualize convergence behavior, which showed gradual improvement and eventual stabilization near epoch 18.

(3) Sampling Rate Mismatch (24 kHz vs 16 kHz)

The F5-TTS base model generates 24 kHz waveforms, while most reference-based metrics (e.g., PESQ, STOI, SISDR) are implemented for 16 kHz signals.

To ensure metric compatibility, all generated audios used for reference-based evaluation were resampled to 16 kHz, while DNSMOS and SRMR were evaluated directly at 24 kHz. This resampling step, although necessary, introduces minor information loss, which may artificially lower absolute scores. However, since both base and fine-tuned models were resampled under identical conditions, the relative improvements remain valid.

(4) Unavailable Metrics (NISQA, DISTILL_MOS)

Attempts to compute NISQA and DISTILL_MOS failed due to environment incompatibility in the provided SpeechScore package. These metrics were excluded to maintain experimental stability. Instead, DNSMOS and SRMR were used as non-intrusive substitutes, both of which are widely accepted in TTS evaluation.

(5) Low Absolute but Consistent Relative Metrics

Although the fine-tuned model showed clear relative improvements across metrics (+4.8 dB SISDR), the absolute values of intrusive metrics ($\text{PESQ} \approx 1.06$, $\text{STOI} \approx 0.10$) remained low. This phenomenon is largely attributed to temporal misalignment between generated and reference audios. Strict intrusive metrics like PESQ and STOI penalize even small temporal or tonal mismatches, leading to systematically low absolute values.

Therefore, while the absolute metric values appear small, the relative deltas (Δ) remain reliable indicators of improvement. Subjective listening confirms that the fine-tuned model produces perceptibly clearer and more natural speech despite these numerical artifacts.

(6) Dataset Constraints

The personalized dataset (~1 hour of high-quality Paimon speech) was clear and consistent, but its limited duration may restrict the model's ability to fully learn speaker timbre and prosody diversity. Longer or multi-condition fine-tuning could yield further gains.

AI Usage: I use AI to debug my code, generate some code, ask for the meaning and the mechanism of different metrics in speechscore, ask for how nfe and cfg work, and help me refine my report writing. I understand AI can be used as a tool and is not a substitute for doing my own work.